Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C046 **KSS1** 4 genes / 3 species 3 direct pairs	**Y1 KSS1** GenAge 397 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 17662940	**W1 mpk-1** GenAge 961 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18059442, 20624915	**F1 Mpk2** GenAge 1832 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 22014527 **F2 p38b** GenAge 1831 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 22014527	No gene in this candidate network
Direct pairs: W1–Y1 (n:m); F1–Y1 (n:m); F2–Y1 (n:m)				
C099 **MNS1** 3 genes / 3 species 2 direct pairs	**Y1 MNS1** GenAge 1513 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 20657825	**W1 D2030.1** GenAge 912 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 19302370	**F1 alpha-Man-I** GenAge 833 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 19302370	No gene in this candidate network
Direct pairs: W1–F1 (1:1); F1–Y1 (n:1)				
C044 **nac-2** 4 genes / 3 species 3 direct pairs	No gene in this candidate network	**W1 nac-2** GenAge 452 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 14678010 **W2 nac-3** GenAge 453 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 12480943	**F1 Indy** GenAge 366 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 11118146, 17571923 18460334	**M1 Slc13a1** GenAge 825 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 21651971
Direct pairs: W1–F1 (m:1); W2–F1 (m:1); F1–M1 (m:n)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.